

Deepak Rathore

AI Engineer | GenAI Applications | RAG & Agentic Systems | Python Backend

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of production experience building enterprise GenAI applications, RAG pipelines, and agentic workflows across procurement, logistics, healthcare, and HR domains. Delivered measurable business impact including 90% reduction in document review time, 97% reduction in bulk-processing time, and ~40% reduction in LLM API costs.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, Bash

Generative AI & LLMs: OpenAI GPT-4/4o, Google Gemini API, Claude (Anthropic API), Llama, Hugging Face Transformers, prompt engineering, tool calling, hallucination reduction

RAG & Agentic AI: LangChain, LangGraph, LlamaIndex, RAG architecture, Model Context Protocol (MCP), CrewAI, AutoGen, ReAct agents, multi-agent systems, intent classification, semantic search, hybrid search, PyMuPDF, Tesseract OCR

Vector DB & Evaluation: ChromaDB, Pinecone, FAISS, pgvector, MiniLM embeddings, FlashRank reranking, RAGAS

Backend & Databases: FastAPI, Django, Flask, Node.js, REST APIs, async I/O, webhooks, PostgreSQL, connection pooling, JWT Authentication, RBAC

Cloud, DevOps & AI Dev Tools: AWS Bedrock, Vertex AI, Docker, GitHub Actions, GitLab CI/CD, Prometheus, Grafana, Git, Postman, Streamlit, OWASP LLM Top 10, Cursor, GitHub Copilot, Claude Code, Codex CLI, Antigravity

PROFESSIONAL EXPERIENCE

Softude Infotech Pvt. Ltd. | *AI Engineer*

Jun 2025 – Present | Indore, India

- Designed a production RAG procurement chatbot for DCGPAC using LangChain, MiniLM dense retrieval, and GPT-4, supporting 500–1,000 purchase orders daily and automating 100–500 CRM emails weekly across 3 enterprise workflows.
- Built an autonomous ReAct sales intelligence agent for DCGPAC that monitors live web signals for the top 50 clients and routes personalized alerts, helping secure 50% of orders in advance with upfront payment.
- Built an end-to-end OCR and classification pipeline for Medico Healthcare, processing 500+ patient records monthly into PostgreSQL, reducing manual patient-record retrieval time by 90% for hospital staff.
- Re-engineered a bulk upload pipeline for enterprise clients including Havells, Hero, and TVS, cutting 10,000-record processing time from 10–15 minutes to under 30 seconds (97% reduction) using async I/O and connection pooling.
- Built a maritime document intelligence pipeline (OCR, PDF parsing, FlashRank cross-encoder reranking) that cut analyst review time from 3+ hours to under 15 minutes per batch, a 90% reduction.

RA Software | *Python AI Engineer*

Mar 2024 – May 2025 | Jaipur, India – Remote

- Engineered an enterprise HR assistant using LangChain and a custom semantic caching layer (ChromaDB) with cosine-similarity thresholding; slashed API costs by ~40% and latency by 65% while enforcing Pydantic validation to eliminate data hallucinations.
- Designed and scaled high-throughput Django REST APIs managing onboarding, attendance, and payroll modules; optimized PostgreSQL queries and indexing, reducing end-to-end API response times by 30%.
- Architected an asynchronous data worker pipeline using Celery and Redis to handle heavy, text-based performance review data ingestion and automated classification, removing processing bottlenecks.

PROJECTS

- OMNIMIND – Agentic AI Research Platform** (Python, LangChain, LlamaIndex, LangGraph, FastAPI, ChromaDB, MCP): Built a multi-agent research platform utilizing the Model Context Protocol (MCP) with three execution paths (RAG, web-search ReAct agent, direct reasoning) unified under an intelligent query router using confidence scoring for sub-second routing decisions.
- Voice AI Receptionist Backend – Dental Appointment Booking Agent** (Python, VAPI, Google Calendar API, Twilio, Firestore, Railway): Built a multi-service backend handling webhook-driven multi-turn conversation state across calls, with real Google Calendar booking, Twilio SMS confirmations, Firestore conversation logging, and an admin REST API; designed for multi-tenant scalability and deployed to a publicly reachable live URL.
- ForgeOps – Multi-Agent Engineering Intelligence Platform** (CrewAI, AutoGen, MCP, Claude API, Next.js, Django): Built three MCP servers exposing repo, test-execution, and codebase-RAG tools to LLM clients, and a multi-agent PR review system combining a sequential CrewAI pipeline with an AutoGen adversarial critique loop. Built a full-stack review dashboard with JWT/RBAC access control via a Django admin panel.

EDUCATION

B.Tech – Computer Science & Engineering, Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal

2021 – 2025 | CGPA: 7.6

Relevant Coursework: Data Structures & Algorithms, Machine Learning Fundamentals, Database Management Systems (DBMS)